

Project Design Phase

Proposed Solution

Date	29/09/2026
Team ID	SWTID-2026-4654
Project Name	AI StudyBuddy - AI assistant for smarter learning and study support
Maximum Marks	4 Marks

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Students spend too much time manually summarizing lengthy material and preparing flashcards, quizzes and study schedules, and have no single secure place for AI-generated learning resources.
2	Idea / Solution description	AI StudyBuddy is a RESTful backend built with Node.js, Express.js, MongoDB and Mongoose. Authenticated users (JWT, bcryptjs, Student/Admin roles) upload .txt, .md or .pdf material and the Google Gemini API generates summaries, flashcards, multiple-choice quizzes and personalized study plans, which are stored for later retrieval.
3	Novelty / Uniqueness	Content is generated dynamically by Gemini from the student's own material rather than fixed templates, and one platform combines upload, summary, flashcards, quiz, scheduling and history with a modular AI service layer.
4	Social Impact / Customer Satisfaction	Cuts manual preparation time, promotes active recall and self-assessment, supports personalized learning and lowers exam stress for students.
5	Business Model (Revenue Model)	Freemium: free basic generation with paid premium tiers for higher AI usage, larger uploads and advanced study plans; institutional licenses for colleges and coaching centres.
6	Scalability of the Solution	Modular MVC structure and stateless JWT authentication allow horizontal scaling; MongoDB Atlas can scale storage; the AI service layer is isolated so models can be upgraded or new features added with minimal changes.